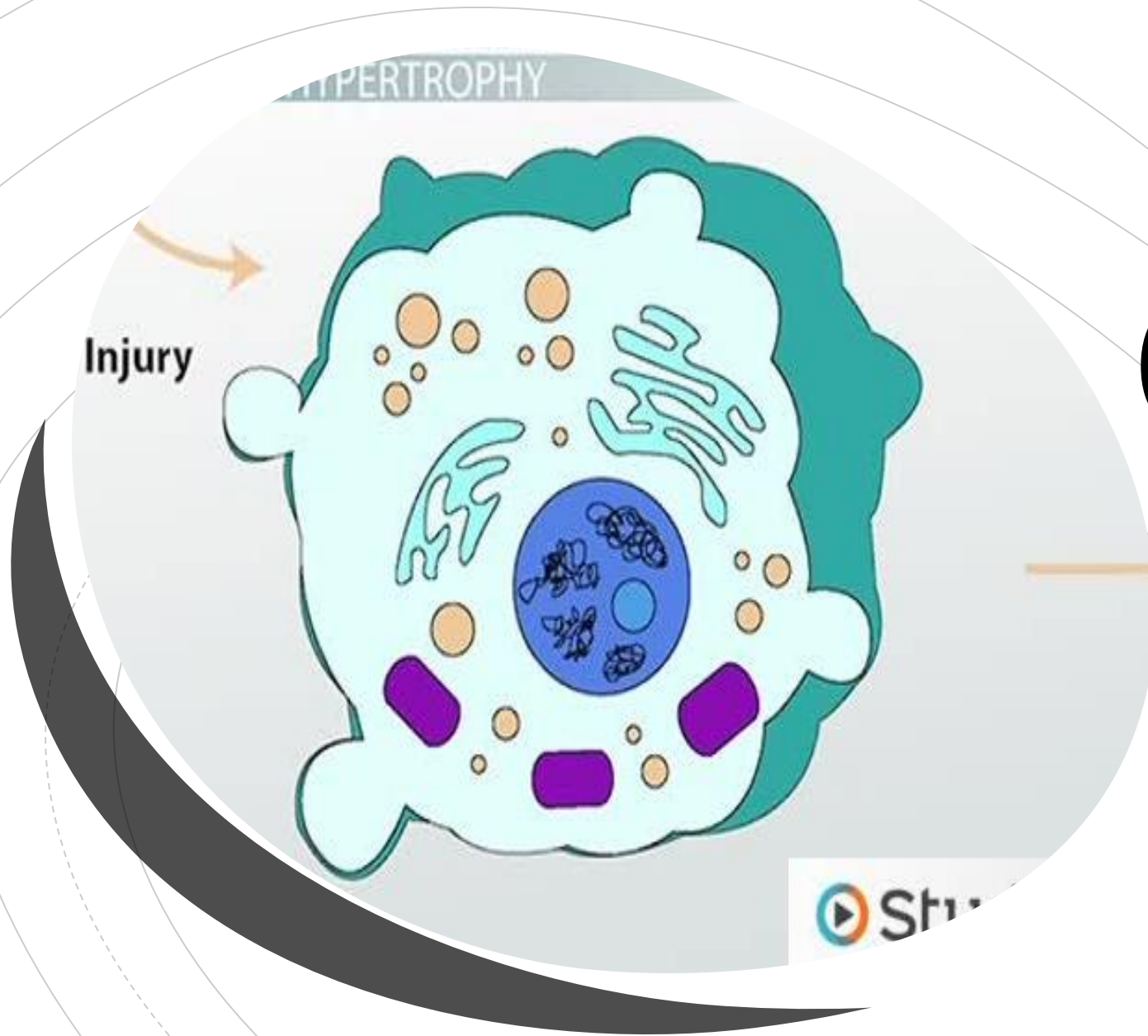




Cell Injury

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MSc, MD

ILOs

- 1- Identify causes and mechanisms of cell injury
2. Know different types of cell injury
3. Identify gross and microscopic picture of reversible cell injury.
 1. Identify types of irreversible cell injury
 2. Identify gross and microscopic picture of irreversible cell injury.
 3. Determine morphological features of necrosis.

NORMAL CELL
(homeostasis)

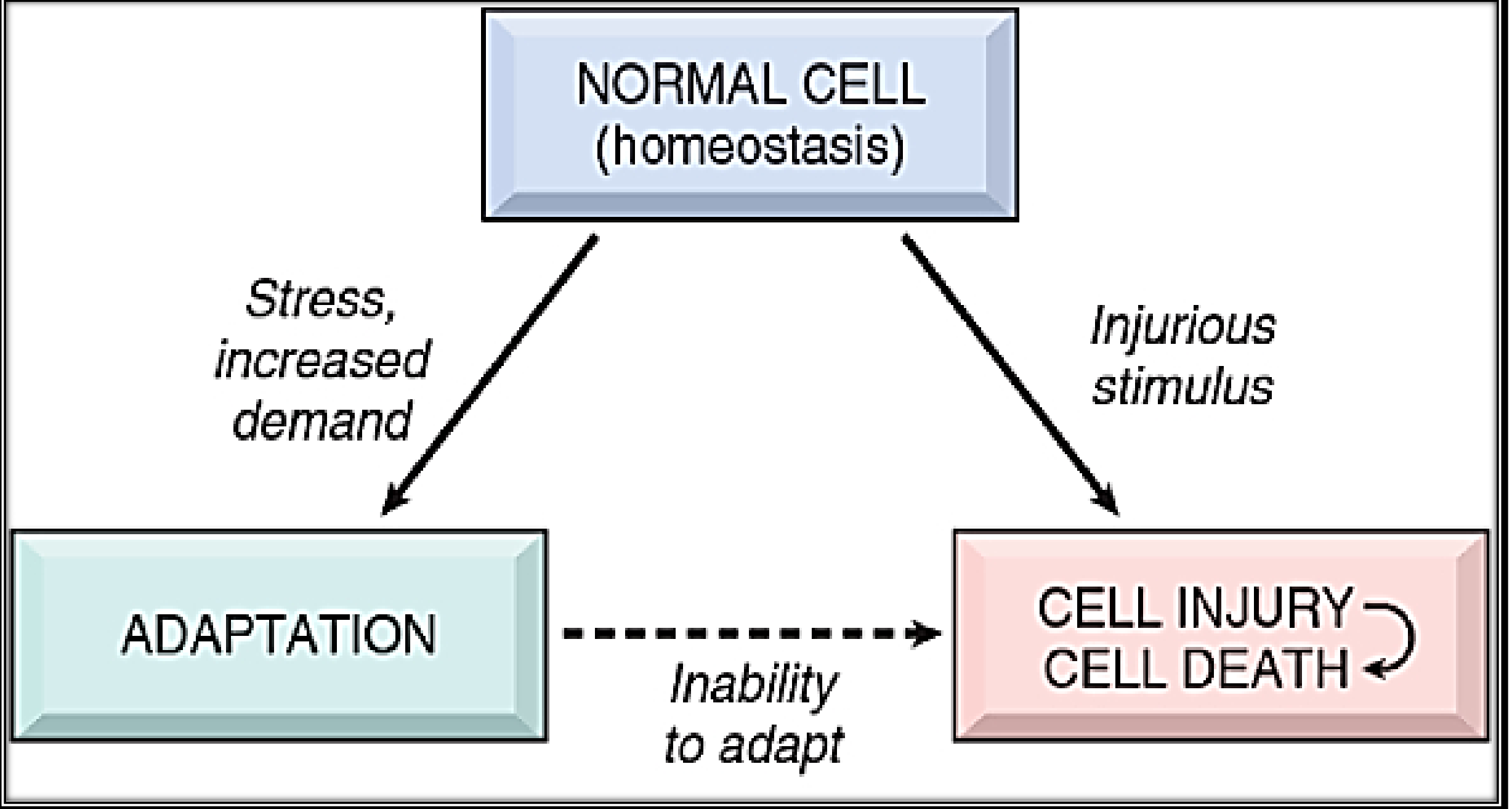
*Stress,
increased
demand*

*Injurious
stimulus*

ADAPTATION

*Inability
to adapt*

CELL INJURY
CELL DEATH



Causes of cell injury:

1- Hypoxia: (Decrease oxygen supply) is caused by:

- a. Ischemia e.g. arterial occlusion and atherosclerosis
- b. Inadequate oxygenation e.g. cardiac and pulmonary disease
- c. Decreased oxygen carrying capacity of the blood e.g. anaemia and carbon monoxide poisoning.

2-Infectious agents: Viruses, bacteria, fungi and parasites.

3- Physical agents: Trauma, heat, cold, irradiation and electric shock.

4- Chemical agents & Drugs: Acids, alkalies, animal and plant poisons, certain therapeutic drugs.

5-Immunological reactions: Autoimmune diseases.

6-Nutritional disturbances.

7-Genetic & Enzymatic disorders.

How injurious agents lead to cell injury ?????



Mechanisms of cell injury

- **ATP depletion:** mainly caused by hypoxia
- **Defects in membrane permeability**
- **Formation of oxygen-derived free radicals:** as superoxide, hydrogen peroxide and hydroxyl radicals.

Mechanisms of cell injury

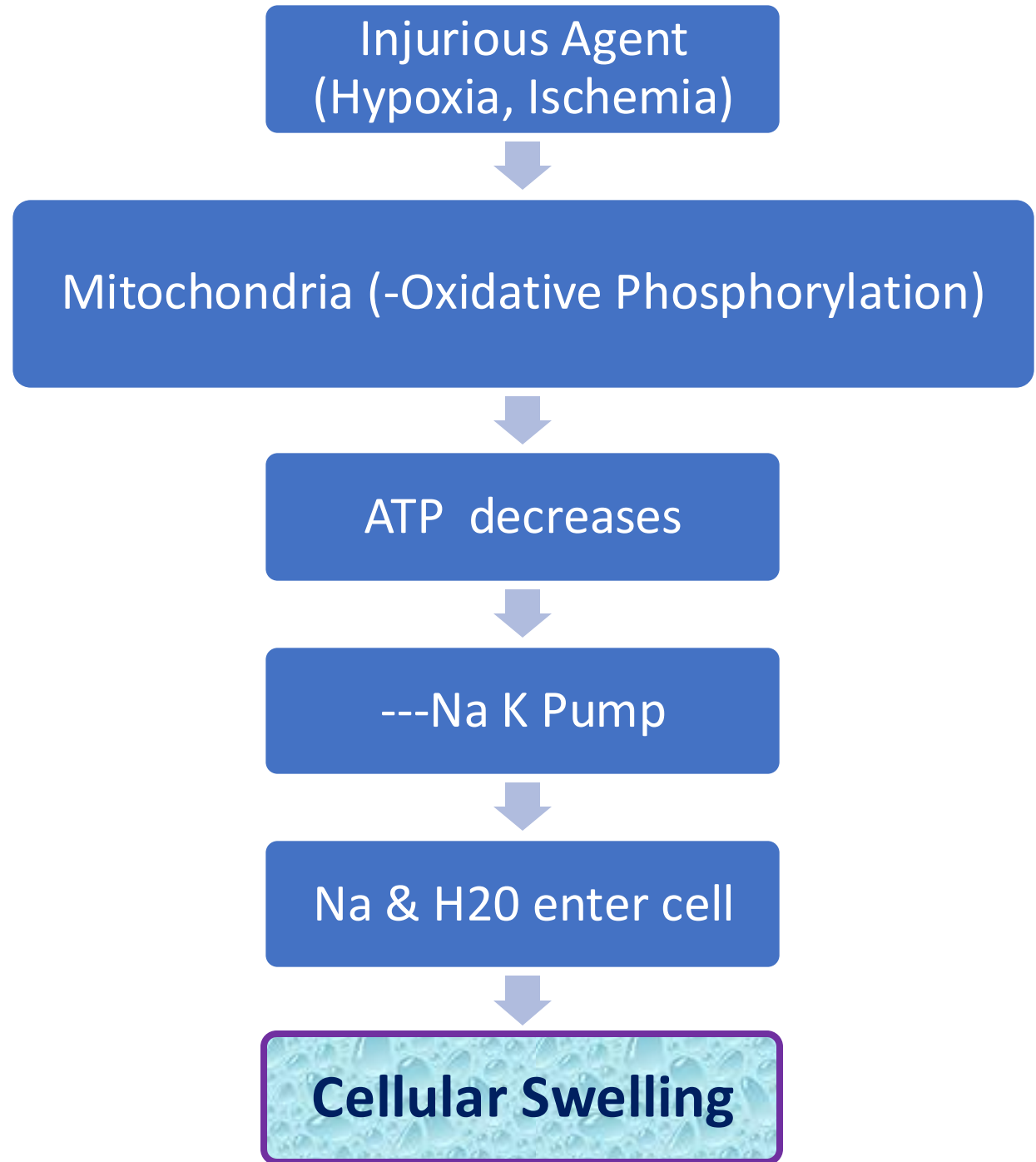
- **Increases intracellular calcium:**

The source of Ca^{2+} is influx from plasma membrane or release from mitochondria and endoplasmic reticulum.

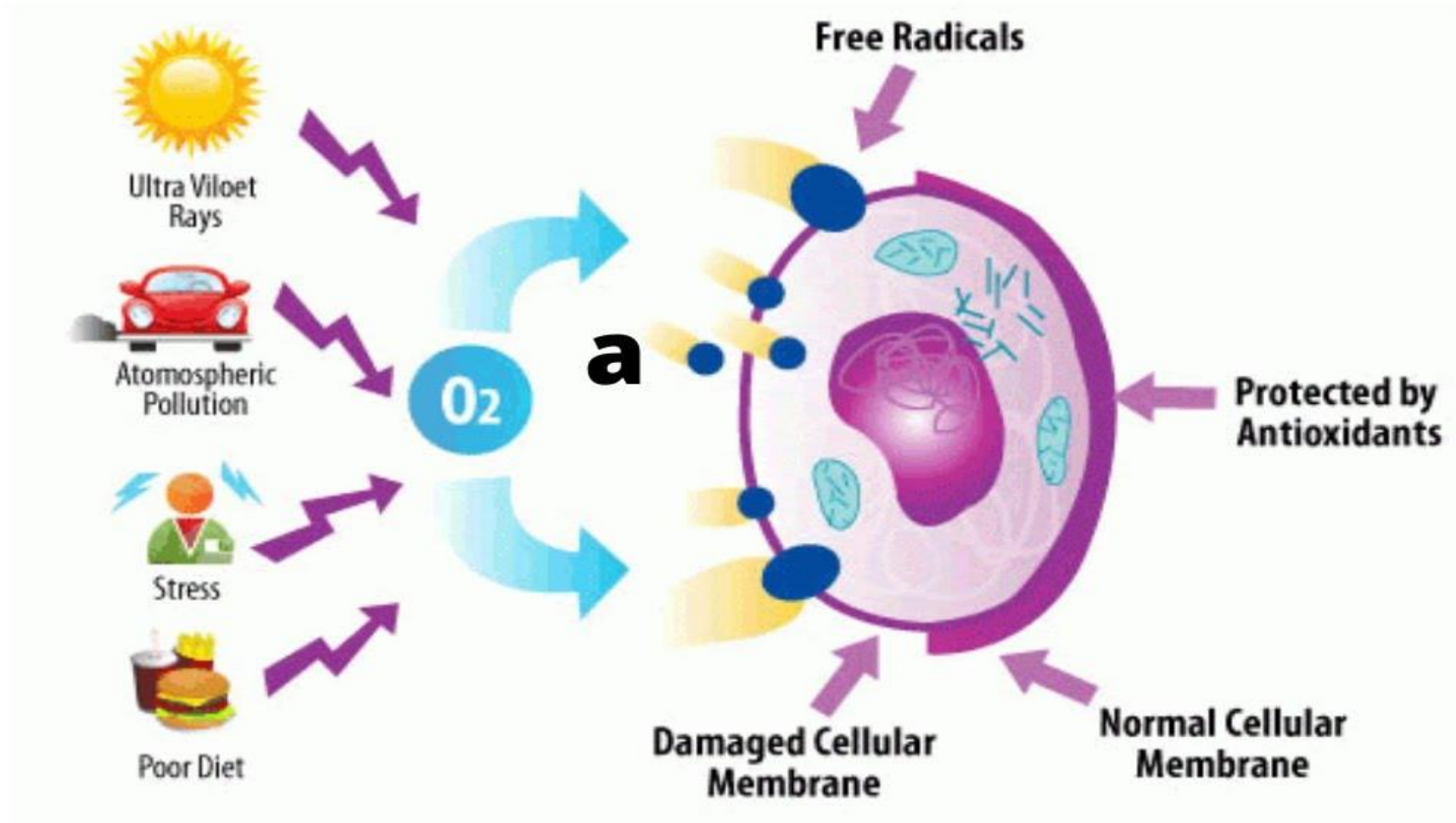
Increased cytosolic Ca^{2+} activates a number of enzymes:

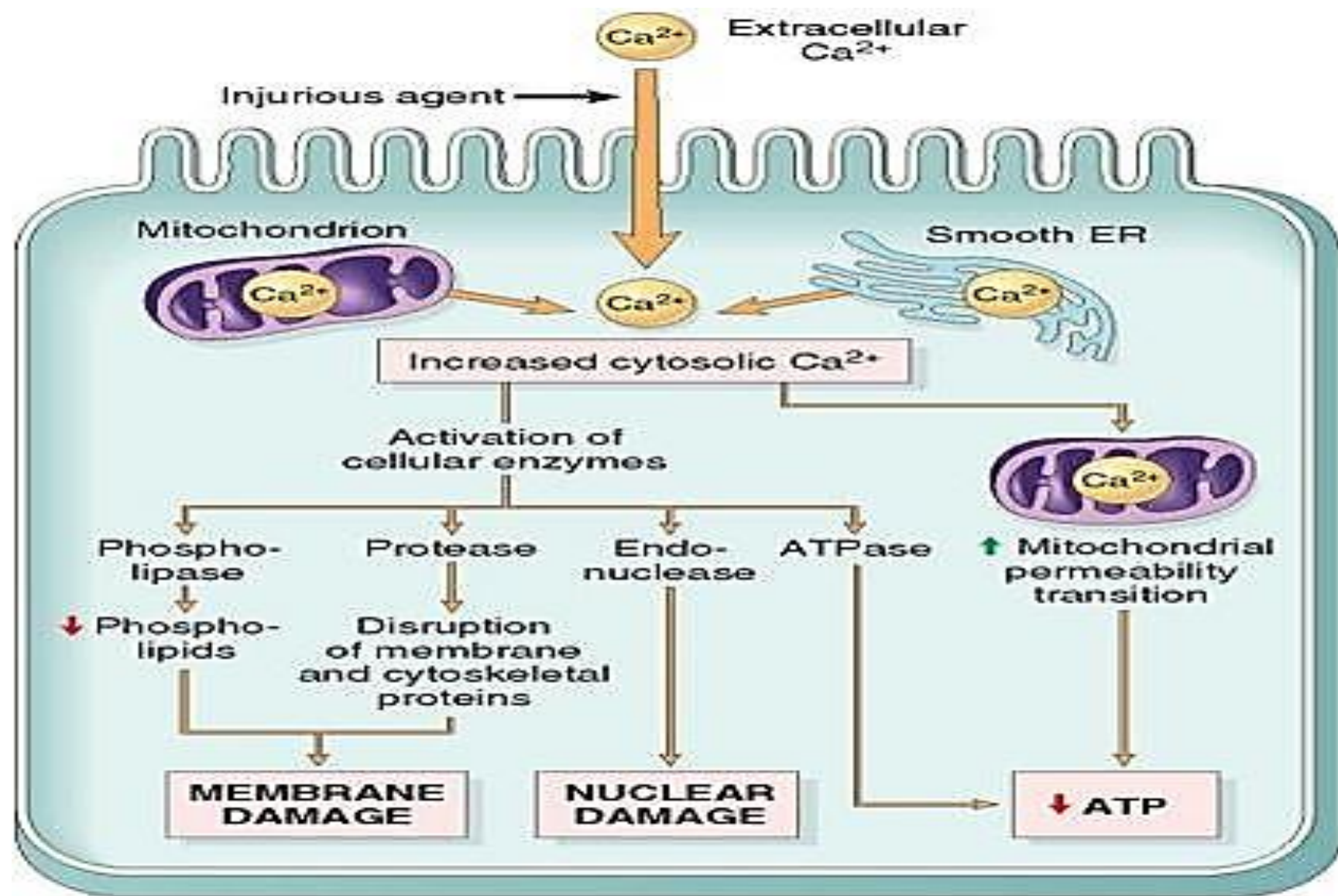
- **A- Phospholipases** degrade membrane phospholipids
- **B- Proteases** break down both membrane and cytoskeletal proteins
- **C- Endonucleases** DNA and chromatin fragmentation
- **D- ATPases** thereby hastening ATP depletion

ATP Depletion Pathogenesis



WHAT ARE **FREE RADICALS**, HOW THEY CAUSE CELL INJURY



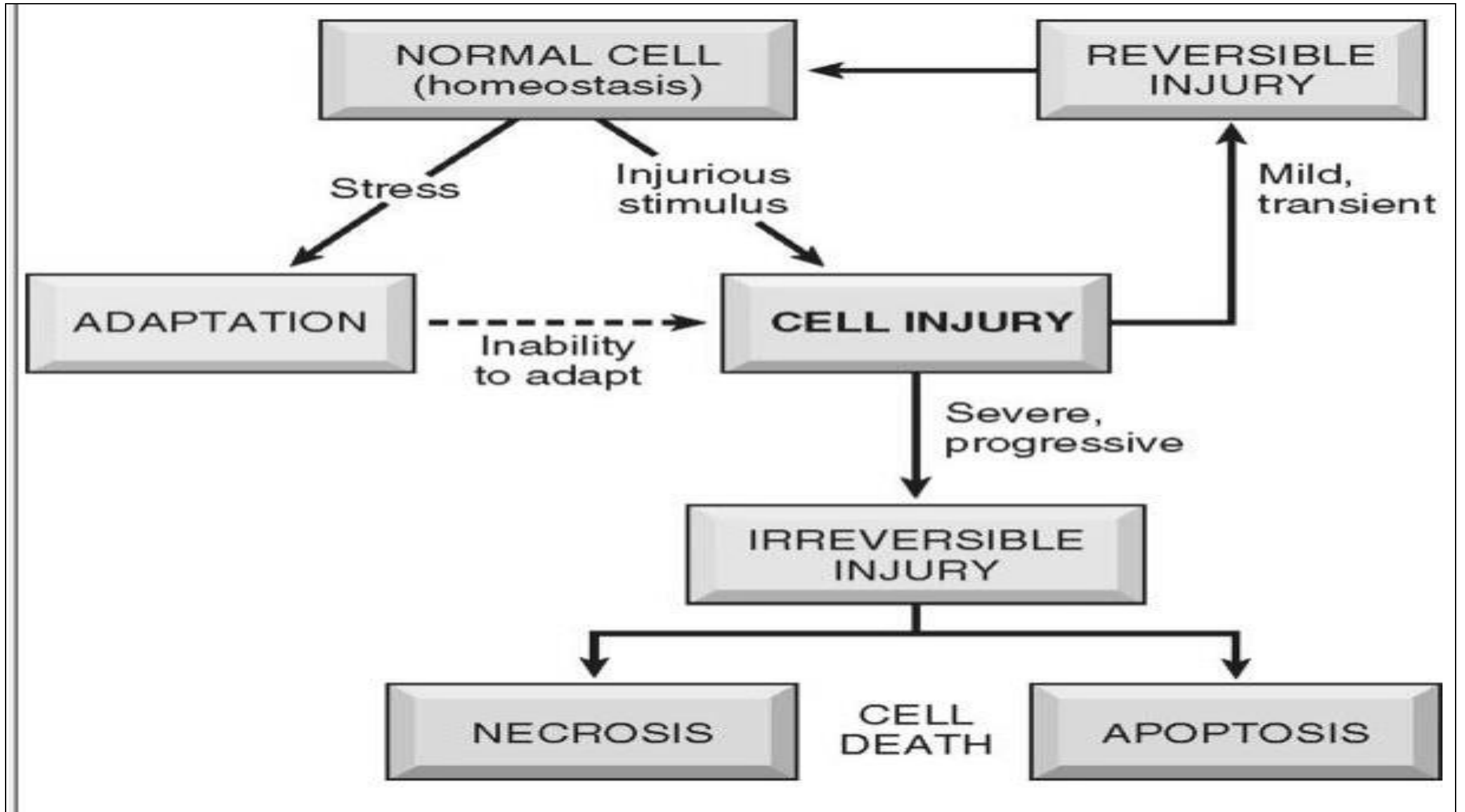


Results of cell injury

Type of injured cells

Duration

Severity of irritant



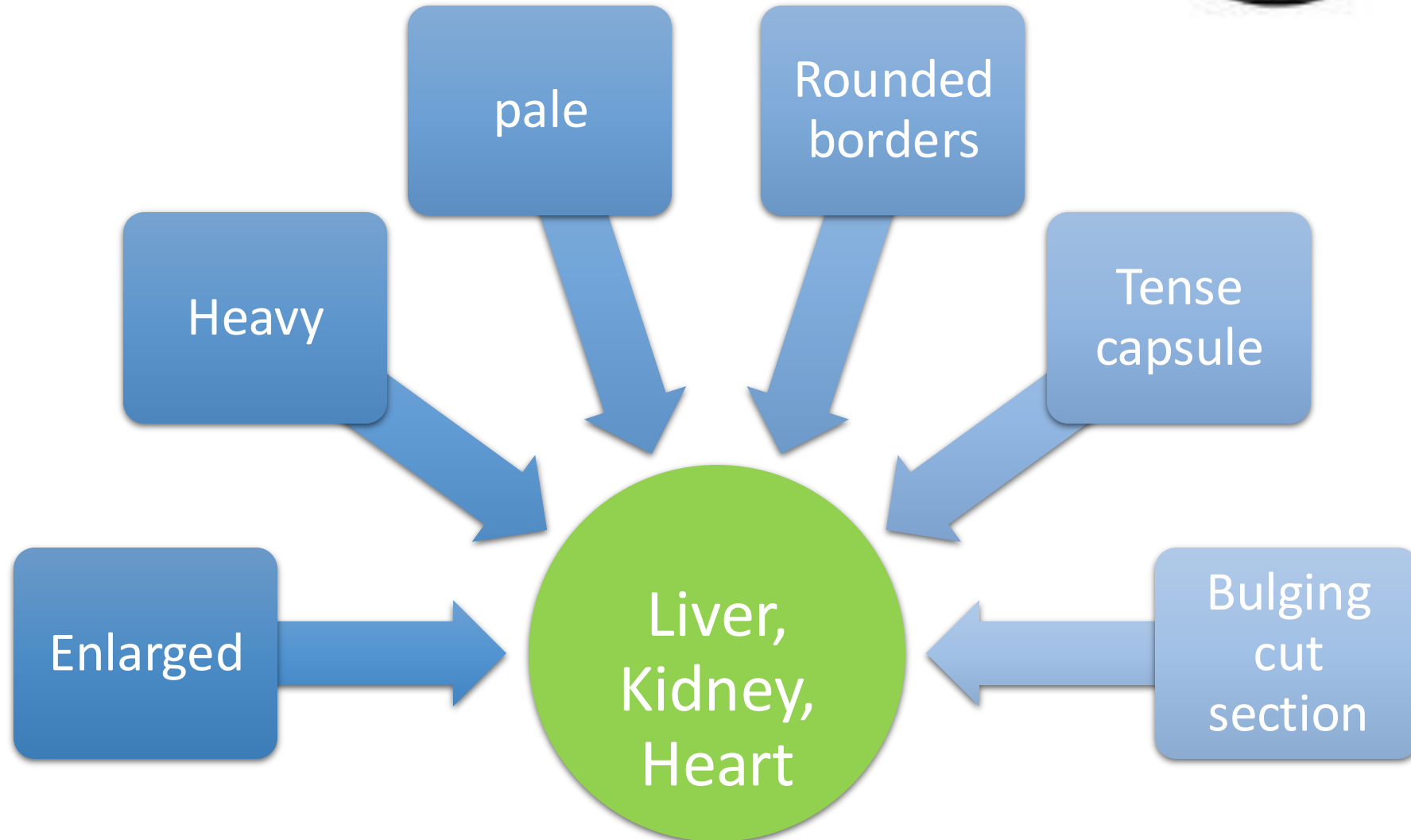
(I) Reversible Cell Injury (Degeneration)

Cloudy swelling:

- **Pathological accumulation of water.**
- **Reversible cell injury characterized morphologically by swelling of the cells (due to water accumulation) and granularity of the cytoplasm.**



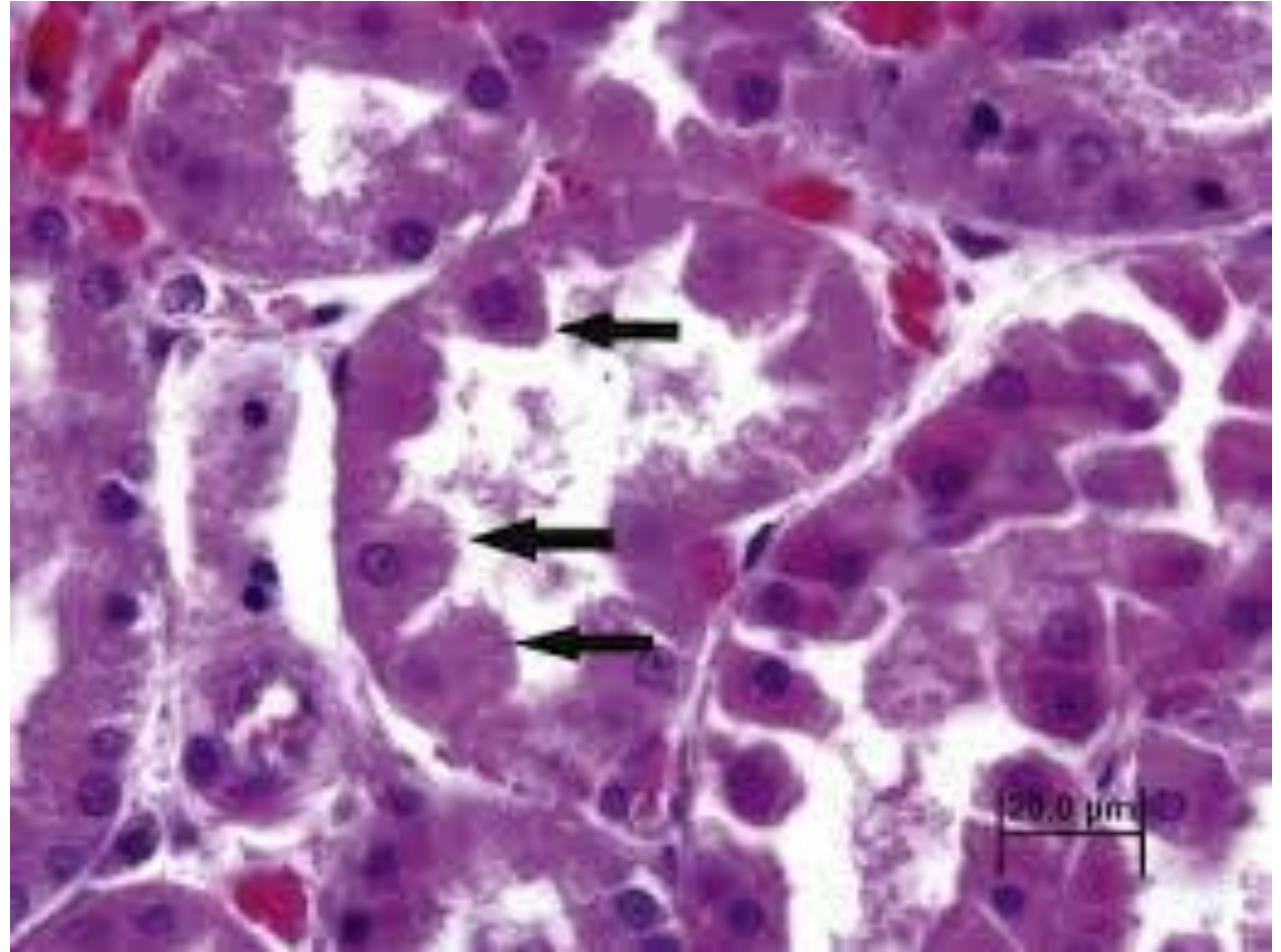
Gross Picture:





Microscopic picture

- Cells :Swollen
- Cytoplasm: Fine red granules
- Nucleus: Normal
- Capillaries between cells: Compressed.



Fate:

If the injury stops
→ Reversible
(cell returns to
normal).

If the injury
continues →
Hydropic
swelling

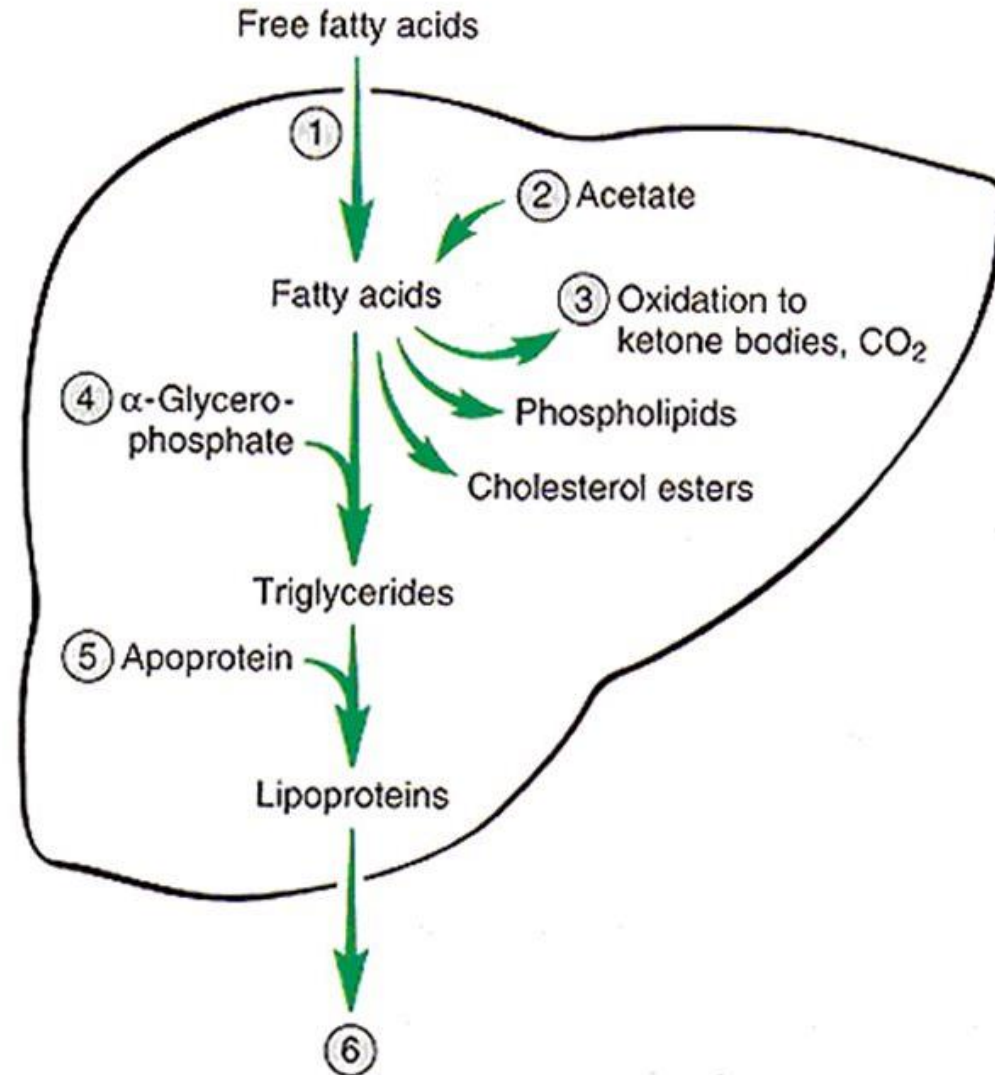
Fatty change

Definition:

Pathological accumulation of excess neutral fat in parenchymatous cells



Mechanism of fat metabolism:



Fatty change in liver cells are also caused by:

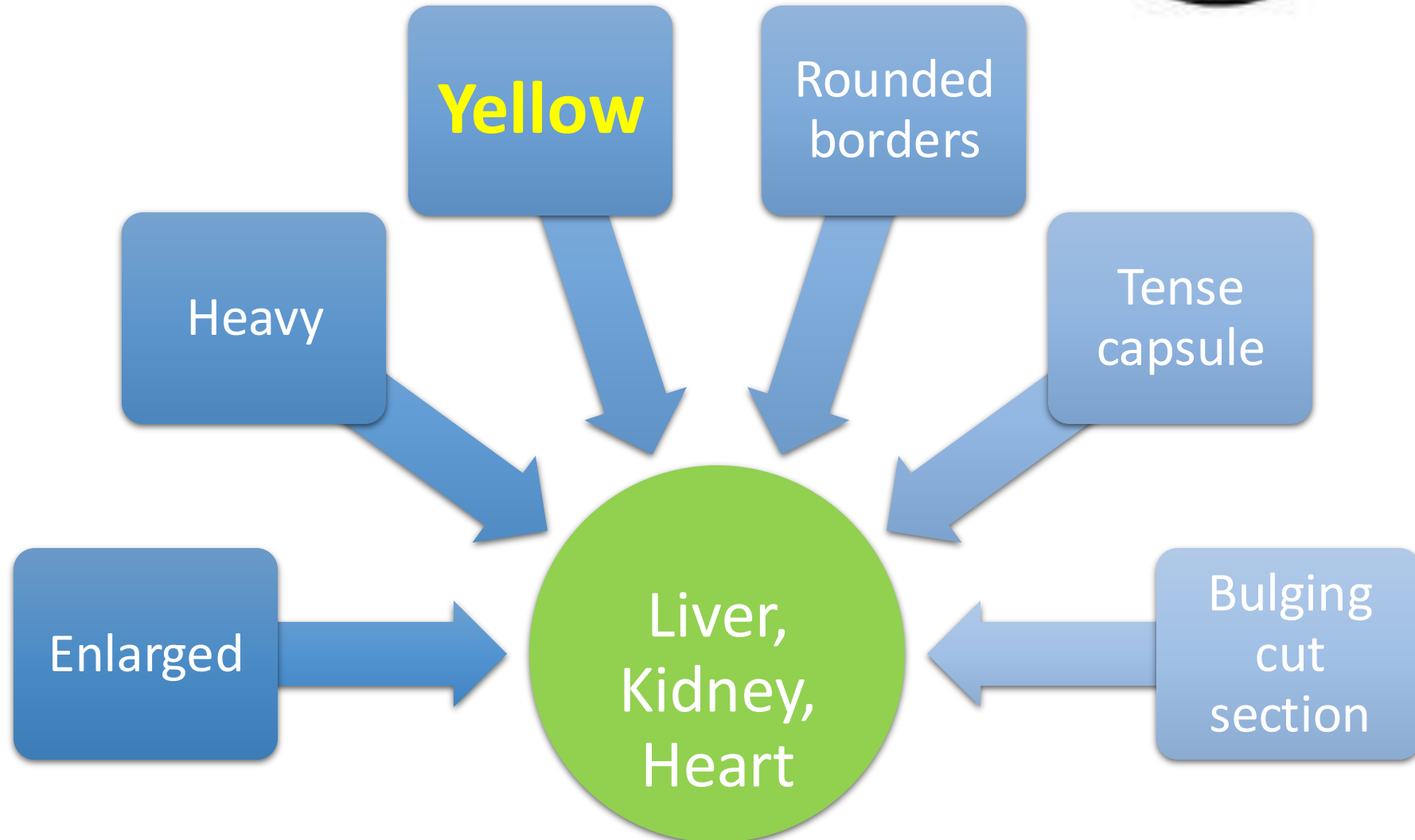
a) Excess fat brought to the liver cells as in:

- **Excess intake of fat and carbohydrate**
- **Starvation with excess mobilization of fat from stores**

a) Diseases of the liver cells as viral hepatitis. The diseased cells cannot metabolized the normal amount of fat reaching it.

b) Deficiency of lipotropic factors as choline and methionine. These are necessary for transformation of neutral fat to choline containing phospholipids as lecithin (the form of fat which most of the body cells can utilize)

Gross Picture:

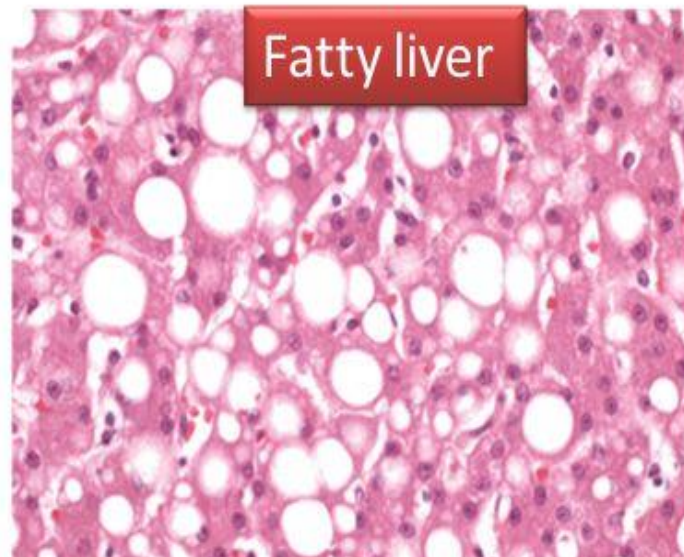
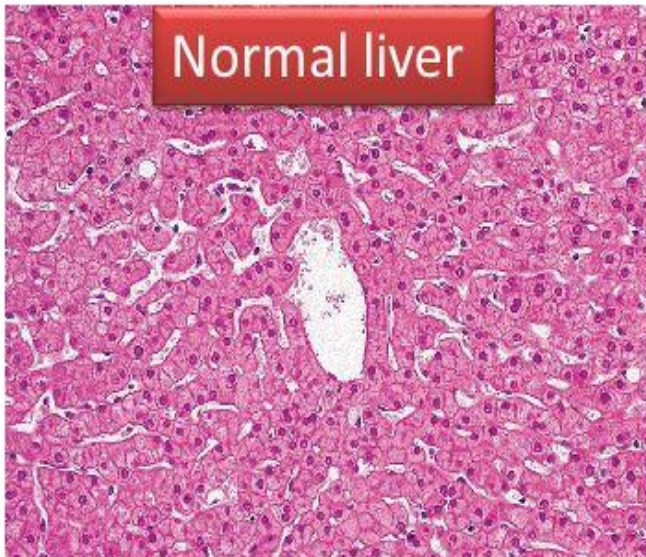




2. Fatty change

- Lipid vacuoles in the cytoplasm.
- In cells participating in fat metabolism (e.g., hepatocytes and myocardial cells).

Signet ring appearance



(II) Irreversible Cell Injury

Necrosis

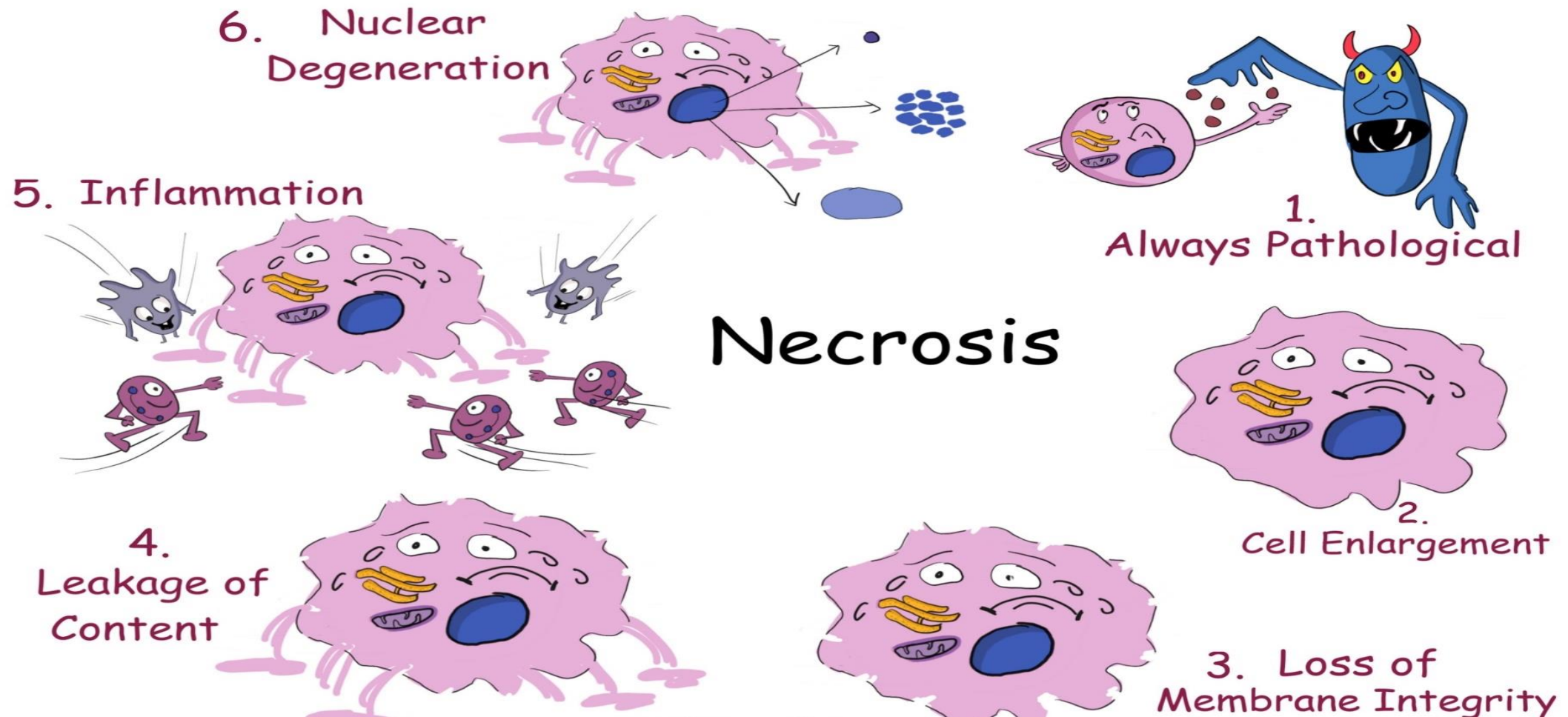
Definition

- Local death of a large groups of cells or tissues within the living body. It occurs either directly or follows reversible injury.

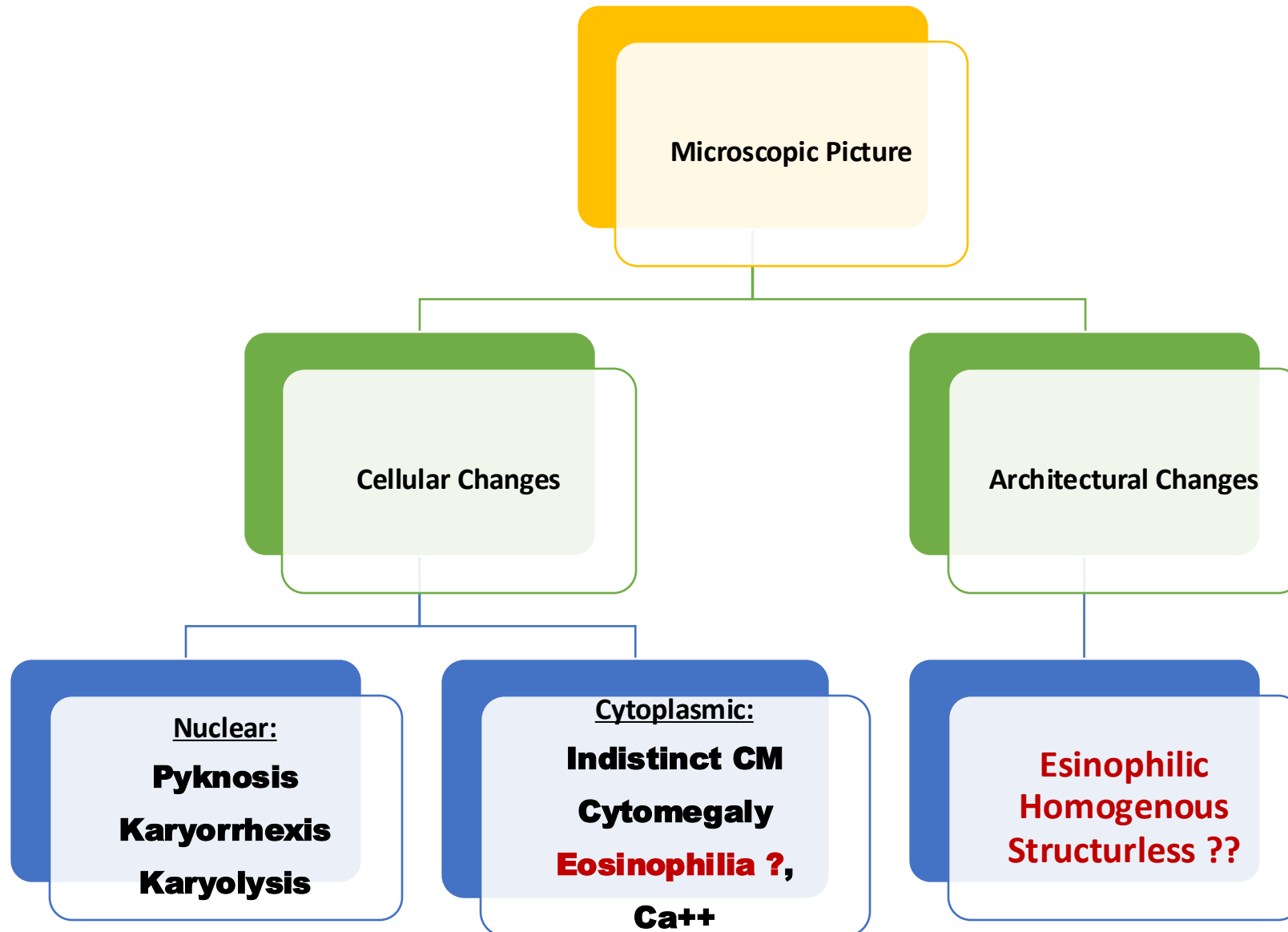
Causes

- Physiological: none

Pathogenesis:



Morphologic changes:



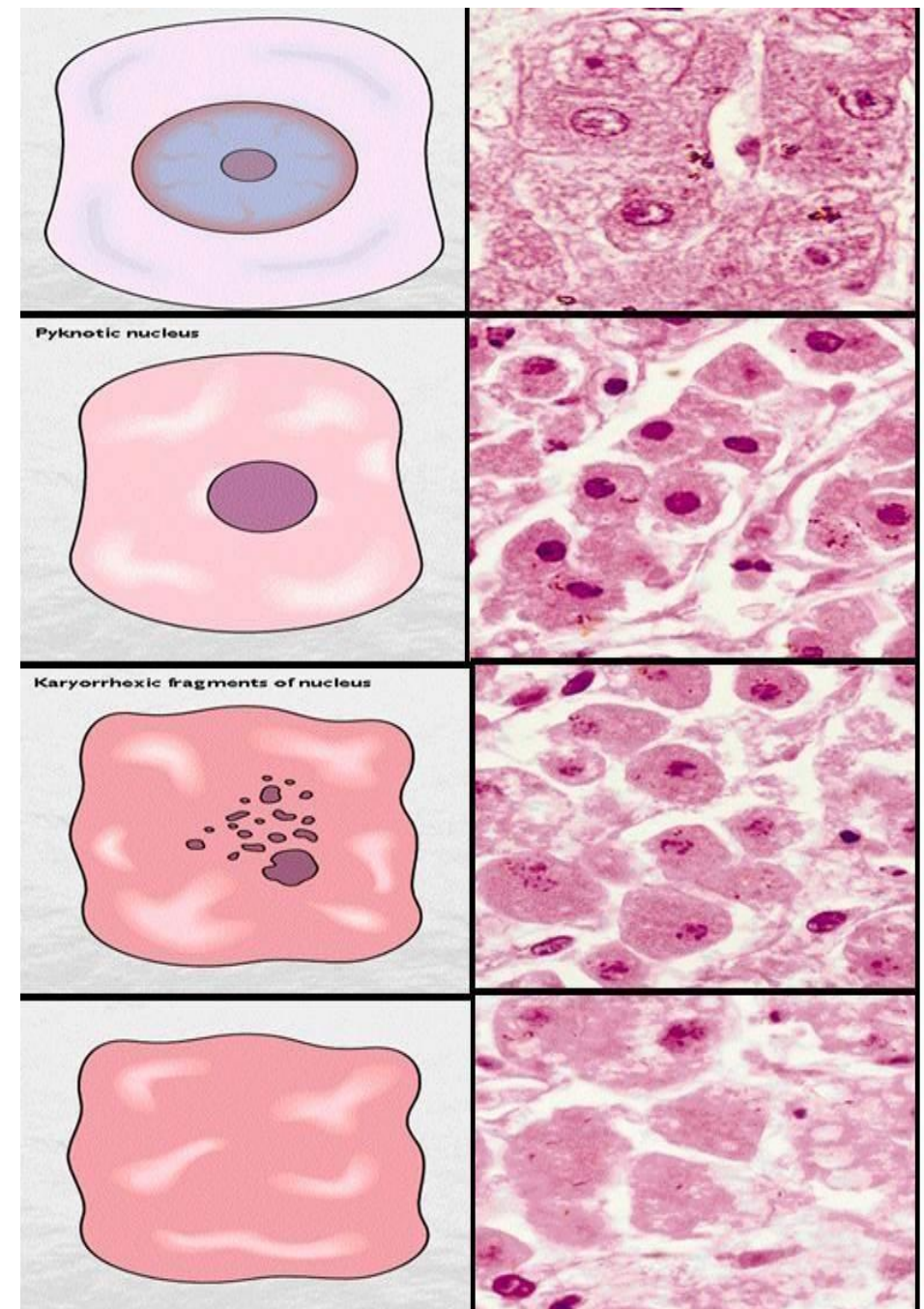
- **Gross picture**

- Opaque, white or yellow surrounded by red area due to inflammatory hyperemia

- **Microscopic picture**

Nucleus

- ***Pyknosis:** Nuclear shrinkage with dense chromatin and increased basophilia.
- ***Karyorrhexis:** The pyknotic nucleus undergoes fragmentation (The nucleus breaks up to multiple small fragments)
- ***Karyolysis:** The nucleus dissolve and fail to take the stain due to chromatin hydrolysis.



A

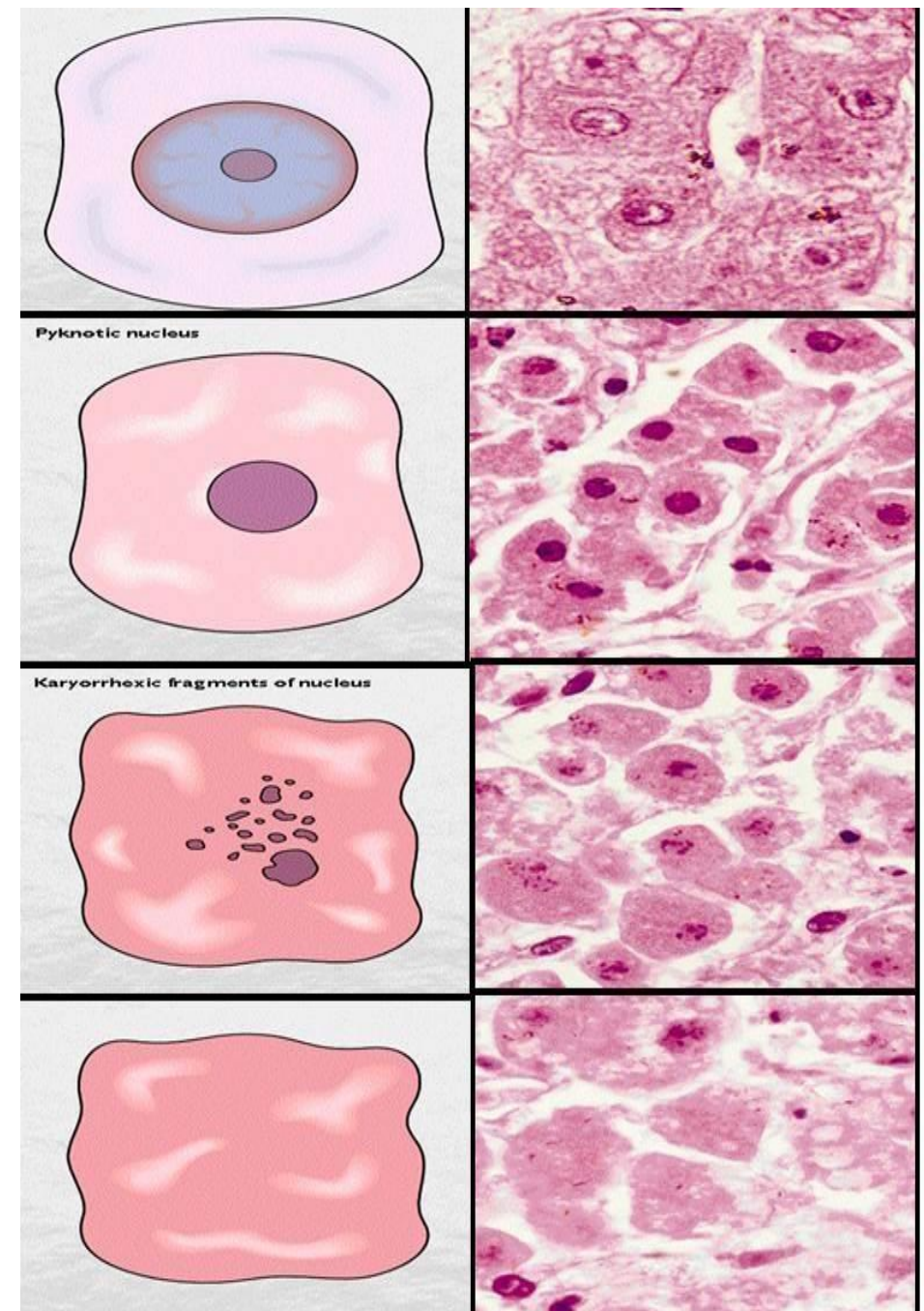
B

C

D

Cytoplasm

- *Cytomegaly (cells are swollen).
- *Cytoplasmic eosinophilia in H&E stained sections: attributable in part to the loss of cytoplasmic RNA (which binds the blue dye, hematoxylin) and in part to denatured cytoplasmic proteins (which bind the red dye, eosin).
- *Glassy homogeneous appearance mainly as a result of the loss of glycogen particles.
- *Indistinct cell membrane.
- *Ca²⁺ deposition.

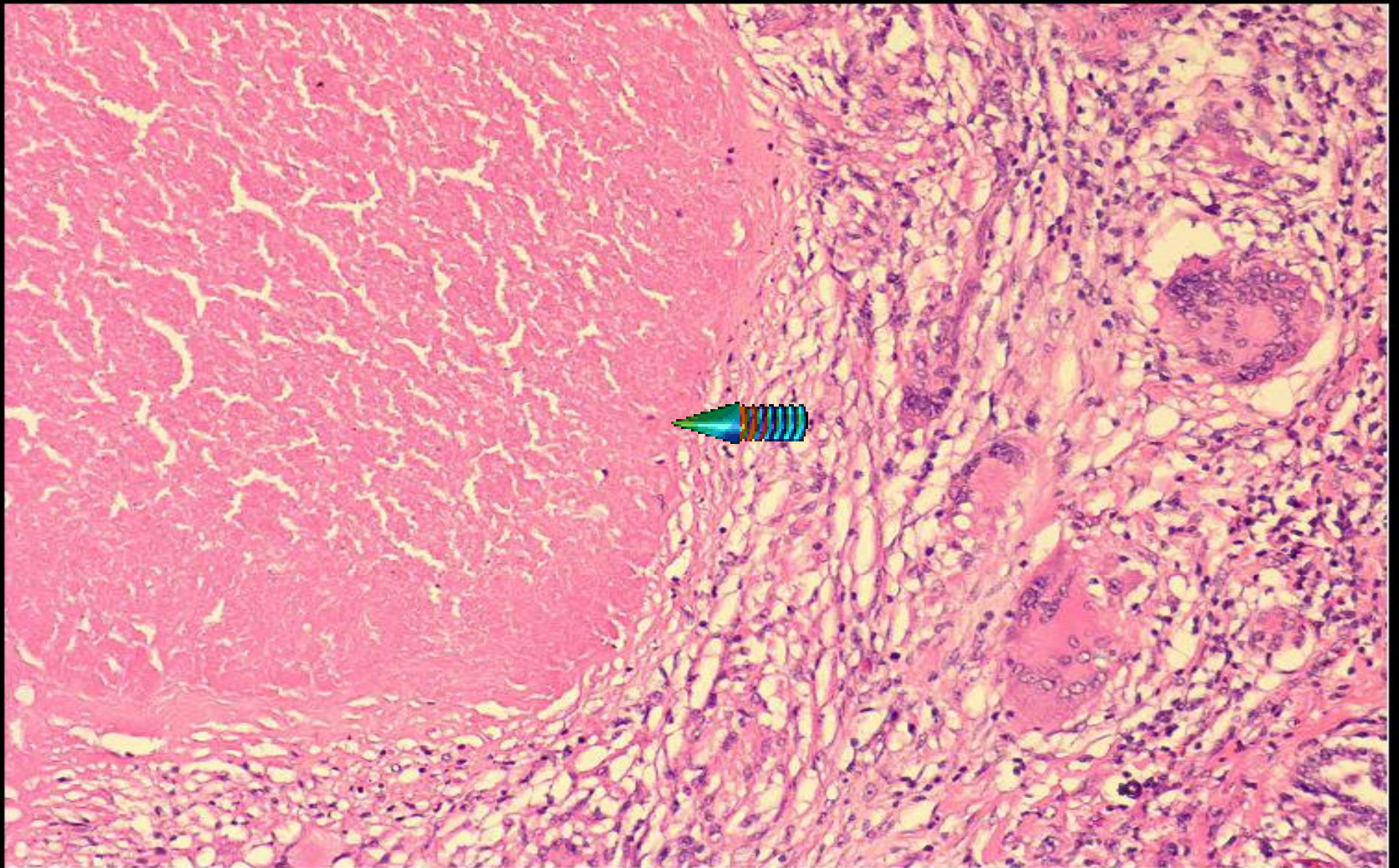


A

B

C

D



Thank
You